

ASRITH REDDY VELIREDDY

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience architecting, fine-tuning, and deploying production-grade Machine Learning models, Computer Vision systems, and Generative AI pipelines. Proficient in LLM fine-tuning (LoRA/QLoRA), Retrieval-Augmented Generation (RAG) architectures, and Agentic workflows using PyTorch, LangChain, and Hugging Face. Demonstrated track record in containerizing low-latency RESTful microservices with FastAPI and Docker, orchestrating distributed inference, and deploying scalable solutions across AWS infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, R

AI/ML & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLOv4, SSD-MobileNet, ResNet-50, XGBoost, LightGBM

Generative AI & LLMs: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, LoRA/QLoRA, vLLM, TensorRT-LLM, Prompt Engineering, RAG Architectures, Vector DBs (Pinecone, FAISS, ChromaDB), NeMo Guardrails

MLOps, Cloud & DevOps: FastAPI, Docker, Kubernetes (K8s), MLflow, AWS (SageMaker, EC2, S3), GitHub Actions, CI/CD, Prometheus, Triton Inference Server

Data Processing & Storage: PostgreSQL, MySQL, Pandas, NumPy, Apache Spark

WORK EXPERIENCE

AI/ML Engineering Intern (CPT) | LeewayHertz | East Coast, USA

Jan 2025 – Dec 2025

- Architected and deployed an enterprise RAG pipeline across 100,000+ unstructured corporate documents, improving search retrieval precision by 32% and reducing hallucination rates by 18% utilizing LangChain, OpenAI API, and Pinecone hybrid vector search.
- Fine-tuned open-source LLMs (Llama-3-8B and Mistral-7B) via Hugging Face using LoRA/QLoRA, reducing GPU memory footprint by 65% while increasing domain-specific intent classification accuracy by 14%.
- Engineered autonomous multi-agent reasoning workflows using LangGraph and dynamic function calling, automating multi-step document audits and reducing manual processing time by 40%.
- Deployed low-latency RESTful ML inference microservices using FastAPI and Docker on AWS SageMaker, maintaining sub-120ms response latencies under concurrent workloads of 250+ requests per second.

Associate AI/ML Engineer | Vue.ai (Mad Street Den) | Chennai, India

Jun 2023 – Dec 2023

- Engineered automated document and visual extraction pipelines utilizing YOLOv4 and SSD-MobileNet architectures with custom OCR post-processing, increasing overall data parsing accuracy by 12%.
- Curated, annotated, and sanitized custom vision datasets comprising 10,000+ images; applied dynamic preprocessing and OpenCV-driven augmentation techniques to eliminate data drift and reduce model overfitting.
- Containerized machine learning inference workloads using Docker and architected low-latency RESTful APIs via FastAPI, maintaining consistent sub-150ms response times for real-time document analysis.
- Implemented automated pattern-matching algorithms and custom validation filters to sanitize raw optical outputs, reducing downstream verification workloads by 25%.

Data Scientist | SRET | Chennai, India

Oct 2022 – May 2023

- Architected and evaluated automated license plate recognition models utilizing SSD-MobileNet and YOLO architectures with integrated OCR engines, achieving an 84% detection accuracy across varying real-world lighting and motion conditions.
- Engineered end-to-end image preprocessing and augmentation pipelines using Python, NumPy, and OpenCV on a curated dataset of 5,000+ custom annotated images, mitigating model overfitting and reducing artifact noise by 30%.
- Designed custom filtering algorithms and heuristic pattern recognition logic for optical character recognition post-processing, raising raw character extraction precision by 9.5%.
- Optimized model inference latency and memory footprints via bounding box threshold tuning and image normalization techniques, achieving sub-100ms per-frame processing speeds for real-time edge testing.

Data Analyst Intern | The Sparks Foundation | Chennai, India

Apr 2021 – Jun 2021

- Engineered optimized SQL queries and Pandas data-transformation scripts to aggregate multi-source structured datasets, reducing query execution time by 25% for regular reporting cycles.
- Presented quantitative risk-assessment findings and predictive insights to cross-functional mentors, directly influencing exploratory research direction across 4 distinct geographic focus regions.
- Automated recurring data ingestion and sanitation workflows using Python, eliminating repetitive manual data-cleaning tasks and cutting report turnaround time by 40%.

FEATURED PROJECTS

LLM-Powered Fraud Investigation System | PyTorch, LangChain, FAISS, AWS SageMaker, FastAPI

- Engineered an anomaly detection engine on AWS SageMaker by integrating BERT embeddings with FAISS and Pinecone vector indexing, improving anomalous transaction detection accuracy by 22%.
- Constructed structured JSON output extractors using advanced prompt engineering and Pydantic validation, reducing unstructured financial document processing errors by 30%.

Autonomous Multi-Agent Market Intelligence System | LangGraph, LlamaIndex, vLLM, Docker, FastAPI

- Built an autonomous multi-agent research pipeline using LangGraph to orchestrate concurrent web search, data aggregation, and synthesis tasks, reducing deep-dive industry research time from 4 hours to 3 minutes.
- Optimized local model inference using vLLM and 4-bit AWQ quantization on Mistral-7B, boosting token generation throughput by 3.5x on a single GPU.

Enterprise Multi-Tenant RAG Platform with Guardrails | LlamaIndex, NeMo Guardrails, ChromaDB, FastAPI, Docker

- Architected a production RAG system incorporating semantic chunking, re-ranking (Cohere Rerank), and NeMo Guardrails, preventing prompt injection attacks and reducing out-of-domain hallucinations by 95%.
- Containerized and served the pipeline via FastAPI and Docker with asynchronous request batching, sustaining 50+ concurrent queries with an average latency under 450ms.

Automated Diabetic Retinopathy Severity Grading | PyTorch, ResNet-50, OpenCV, Docker

- Trained a ResNet-50 transfer learning model on Kaggle APTOS datasets to classify retinal disease severity, attaining an 83% multi-class accuracy and a Quadratic Weighted Kappa score of 0.8712.
- Implemented class-balanced loss functions and heavy photometric augmentations using OpenCV, mitigating medical dataset imbalance and boosting minority class sensitivity by 16%.

Real-Time Edge Computer Vision & Tracking Engine | OpenCV, YOLOv8, DeepSORT, FastAPI

- Developed a real-time object detection and multi-object tracking pipeline using YOLOv8 and DeepSORT, achieving 30+ FPS video stream processing with sub-50ms per-frame inference.
- Integrated inference endpoints within a FastAPI microservice, enabling scalable stream ingestion and automated alert triggers for edge devices.

EDUCATION

Master of Science in Computer Science <i>New Jersey Institute of Technology (NJIT), Newark, NJ</i>	Jan 2024 – Dec 2025
B.Tech in Computer Science <i>Sri Ramachandra Engineering and Technology (SRET), India</i>	Aug 2019 – Jun 2023